The Q-Operator and its Adjoint: From Level 3 Operator Traces to Cascade Commutators

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Abstract

We develop two complementary operator-theoretic frameworks for the cylindric Kostka polynomials $Q_{n,c}(q)$.

Part I lifts the construction to Level 3 in the hierarchy of generating functions: the operator trace framework with free parameters. This produces a genuine operator $\hat{Q}_{c,\mathbf{n}}$ on the ring of symmetric functions Λ whose trace recovers the multivariate cylindric Kostka polynomial. The adjoint $\hat{Q}^*_{c,\mathbf{n}}$ corresponds to the reversed-complementary profile, and we show that $\mathrm{tr}(\hat{Q}^*) = \mathrm{tr}(\hat{Q})$: the adjoint is invisible at the polynomial level.

Part II defines a second operator $\mathbf{Q}(z) = \sum Q_m z^m P_m$ that packages the Q-polynomials as coefficients of Pieri multiplication operators. Computing the products $\mathbf{Q}(z)\mathbf{Q}^*(w)$ and $\mathbf{Q}^*(w)\mathbf{Q}(z)$ reveals cross-correlation functions and a cascade commutator structure that couples different depth levels of the cylindric partition.

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Part I
The Level 3 Operator Trace Framework

1 The Level 3 operator

1.1 Setup

Let Λ denote the ring of symmetric functions with Hall inner product $\langle \cdot, \cdot \rangle$ satisfying $\langle s_\lambda, s_\mu \rangle = \delta_{\lambda,\mu}$, or equivalently $\langle h_\lambda, m_\mu \rangle = \delta_{\lambda,\mu}$. The Cauchy identity reads:

$$\sum_\lambda s_\lambda(X)\, s_\lambda(Y) = \prod_{i,j} \frac{1}{1 - x_i y_j} = \Omega[XY]. \tag{1}$$

For rank 3 with profile $c = (c_0, c_1, c_2)$, set $T := d = c_0 + c_1 + c_2$, and let $\pi = (\pi_0, \ldots, \pi_{T-1}) \in \{0, 1\}^T$ be the binary string encoding the profile (with n ones and m zeros, $n + m = T$).

1.2 The vertex operators

Define operators on Λ depending on a formal parameter u:

$$G^1(u) := \Omega[Xu] \quad \text{(Pieri addition: } s_\mu \mapsto \sum_{\lambda \in U(\mu)} s_\lambda \, u^{|\lambda|-|\mu|}\text{)}, \tag{2}$$

$$G^0(u) := \Omega^*[Xu] \quad \text{(Pieri removal: the Hall-adjoint of } G^1(u)\text{)}. \tag{3}$$

In terms of the translation and multiplication operators of [2]:

$$G^1(u) = P(u) = \mathcal{P}(u), \qquad G^0(u) = T(u) = \mathcal{T}(u) = P^*(u). \tag{4}$$

The *degree operator* D_w acts diagonally in the Schur basis:

$$D_w \, s_\lambda = w^{|\lambda|} \, s_\lambda. \tag{5}$$

1.3 The operator product

Definition 1 (Level 3 operator)**.** The *Level 3 operator* for profile π with parameters $\mathbf{u} = (u_0, \ldots, u_{T-1})$ and w is:

$$\hat{O}_c(\mathbf{u}, w) := G^{\pi_0}(u_0) \, G^{\pi_1}(u_1) \cdots G^{\pi_{T-1}}(u_{T-1}) \, D_w \; : \; \Lambda \to \Lambda. \tag{6}$$

The multivariate generating function for cylindric plane partitions is recovered as the trace:

$$\mathrm{GK}_c(\mathbf{z}) = \mathrm{tr}(\hat{O}_c) := \sum_\mu \langle s_\mu, \, \hat{O}_c \, s_\mu \rangle, \tag{7}$$

under the specialisation

$$w = z_0 z_1 z_2, \qquad u_k = \begin{cases} z_0 z_1 \cdots z_{k-1} & \text{if } \pi_k = 1, \\ (z_0 z_1 \cdots z_{k-1})^{-1} & \text{if } \pi_k = 0. \end{cases} \tag{8}$$

Remark 2 (Matrix elements)**.** The trace uses only the *diagonal* matrix elements $\langle s_\mu, \hat{O}_c \, s_\mu \rangle$. The *off-diagonal* elements $\langle s_\lambda, \hat{O}_c \, s_\mu \rangle$ with $\lambda \neq \mu$ count "open" interlacing sequences from μ to λ (without the cylindric periodicity condition $\mu^0 = \mu^T$). The trace enforces periodicity.

2 The adjoint operator

2.1 Adjoint of the building blocks

With respect to the Hall inner product, the vertex operators are mutual adjoints:

$$(G^1(u))^* = G^0(u), \qquad (G^0(u))^* = G^1(u). \tag{9}$$

This is the operator-theoretic content of the Pieri rules: adding a horizontal strip is adjoint to removing one.

The degree operator is self-adjoint:

$$D_w^* = D_w, \tag{10}$$

since $\langle D_w s_\lambda, s_\mu \rangle = w^{|\lambda|} \delta_{\lambda\mu} = \langle s_\lambda, D_w s_\mu \rangle$.

2.2 Adjoint of $\hat{O}_c$

Taking the adjoint of the product (6) reverses the order and replaces each factor by its adjoint:

$$\hat{O}_c^* = D_w \, G^{\bar{\pi}_{T-1}}(u_{T-1}) \, G^{\bar{\pi}_{T-2}}(u_{T-2}) \cdots G^{\bar{\pi}_0}(u_0), \tag{11}$$

where $\bar{\pi}_k := 1 - \pi_k$ is the complementary bit: addition $\leftrightarrow$ removal.

Remark 3 (Profile transformation)**.** The adjoint $\hat{O}_c^*$ is the Level 3 operator for the *reversed-complementary profile* $\bar{\pi}^{\mathrm{rev}} = (\bar{\pi}_{T-1}, \bar{\pi}_{T-2}, \ldots, \bar{\pi}_0)$ with reversed parameters $\mathbf{u}^{\mathrm{rev}} = (u_{T-1}, \ldots, u_0)$, up to the position of D_w.

2.3 Behaviour under the trace

Proposition 4 (Trace invariance)**.** $\mathrm{tr}(\hat{O}_c^*) = \mathrm{tr}(\hat{O}_c)$.

Proof. Since the Hall inner product is symmetric ($\langle f, g\rangle = \langle g, f\rangle$):

$$\mathrm{tr}(\hat{O}_c^*) = \sum_\mu \langle s_\mu, \, \hat{O}_c^* \, s_\mu\rangle = \sum_\mu \langle \hat{O}_c \, s_\mu, \, s_\mu\rangle = \sum_\mu \langle s_\mu, \, \hat{O}_c \, s_\mu\rangle = \mathrm{tr}(\hat{O}_c). \qquad \square$$

Corollary 5 (Profile symmetry)**.** *By cyclic invariance of the trace, $\hat{O}_c^*$ can be rewritten with D_w at the end:*

$$\mathrm{tr}(\hat{O}_c^*) = \mathrm{tr}\big(G^{\bar{\pi}_{T-1}}(u_{T-1}) \cdots G^{\bar{\pi}_0}(u_0) \, D_w\big) = \mathrm{GK}_{\bar{\pi}^{\mathrm{rev}}}(\mathbf{u}^{\mathrm{rev}}, w). \tag{12}$$

Combined with Proposition 4, this gives:

$$\boxed{\mathrm{GK}_\pi(\mathbf{u}, w) = \mathrm{GK}_{\bar{\pi}^{\mathrm{rev}}}(\mathbf{u}^{\mathrm{rev}}, w).} \tag{13}$$

The generating function for profile π equals that for the reversed-complementary profile with reversed parameters.

Remark 6 (The adjoint paradox)**.** The adjoint is **invisible at the trace level**: $\hat{O}_c$ and $\hat{O}_c^*$ produce the same generating function. The adjoint becomes nontrivial only at the level of individual matrix elements $\langle s_\lambda, \hat{O}_c \, s_\mu\rangle$.

3 The multivariate E–H factorisation

3.1 Factorisation at the generating function level

The growth diagram bijection $\psi\colon (\mathrm{IP}, \mathrm{ALCD}(c)) \to \mathrm{CPP}(c)$ decomposes each cylindric plane partition into a base partition γ (contributing $w^{|\gamma|}$ to every slice) and an ALCD $\mathfrak{d}$ (contributing $z_k^{|\mathrm{diag}(k)|_\mathfrak{d}}$ to the k-th slice). Setting $w = z_0 z_1 z_2$:

$$\mathrm{GK}_c(\mathbf{z}) = \frac{1}{\underbrace{(w;\, w)_\infty}_{\displaystyle\sum_\gamma w^{|\gamma|}}} \cdot E_c(\mathbf{z}), \tag{14}$$

where

$$E_c(\mathbf{z}) := \sum_{\mathfrak{d} \in \mathrm{ALCD}(c)} \prod_{k=0}^{T-1} z_k^{|\mathrm{diag}(k)|_\mathfrak{d}} \tag{15}$$

encodes the purely cylindric content.

The *corrected multivariate generating function* is:

$$G_c(\mathbf{z}) := (w;\, w)_\infty \cdot \mathrm{GK}_c(\mathbf{z}) = E_c(\mathbf{z}). \tag{16}$$

3.2 The corrected operator

Definition 7 (Corrected operator)**.** Define the *corrected operator-valued generating function*:

$$\hat{G}_c(\mathbf{z}) := (w;\, w)_\infty \cdot \hat{O}_c(\mathbf{z}), \tag{17}$$

where $(w;\, w)_\infty = \prod_{k\geq 1}(1-w^k)$ is a scalar (depending on $w = z_0 z_1 z_2$) that multiplies the operator.

Its trace recovers the corrected generating function:

$$\operatorname{tr}(\hat{G}_c(\mathbf{z})) = (w;\, w)_\infty \cdot \operatorname{tr}(\hat{O}_c(\mathbf{z})) = G_c(\mathbf{z}).$$

3.3 Multi-index coefficients

Expand the corrected operator in a multi-index power series:

$$\hat{G}_c(\mathbf{z}) = \sum_{\mathbf{n}\in\mathbb{N}^T} \hat{G}_{c,\mathbf{n}}\, z_0^{n_0}\, z_1^{n_1} \cdots z_{T-1}^{n_{T-1}}, \tag{18}$$

where each $\hat{G}_{c,\mathbf{n}} \colon \Lambda \to \Lambda$ is an operator. Note that the Euler factor $(w;\, w)_\infty$ mixes multi-index coefficients: $\hat{G}_{c,\mathbf{n}}$ is *not* simply $(w;\, w)_\infty$ times the $\mathbf{n}$-th coefficient of $\hat{O}_c$, because $(w;\, w)_\infty$ depends on $w = z_0 z_1 z_2$.

The trace of each coefficient gives the corrected multi-index coefficient:

$$g_{\mathbf{n},c} = \operatorname{tr}(\hat{G}_{c,\mathbf{n}}) = [\mathbf{z}^{\mathbf{n}}]\, G_c(\mathbf{z}). \tag{19}$$

4 The Q-operator at Level 3

4.1 Definition

Definition 8 (Q-operator)**.** For a multi-index $\mathbf{n} = (n_0, n_1, \ldots, n_{T-1}) \in \mathbb{N}^T$, profile c with $d \not\equiv 0 \pmod 3$, define the *Q-operator*:

$$\hat{Q}_{c,\mathbf{n}} := \prod_{k=0}^{T-1} (q;q)_{n_k} \cdot \hat{G}_{c,\mathbf{n}} \;:\; \Lambda \to \Lambda, \tag{20}$$

where $q = w = z_0 z_1 \cdots z_{T-1}$ and the $(q;q)_{n_k} = \prod_{j=1}^{n_k}(1-q^j)$ are scalar factors that clear the denominators.

Remark 9. In the rank-3 case, one often restricts to multi-indices of the form $\mathbf{n} = (n_0, n_1, n_2) \in \mathbb{N}^3$ (one index per slice), and the product becomes $\prod_{k=0}^{2}(q;q)_{n_k}$.

4.2 Trace gives the multivariate Q-polynomial

$$Q_{\mathbf{n},c} = \operatorname{tr}(\hat{Q}_{c,\mathbf{n}}) = \prod_k (q;q)_{n_k} \cdot g_{\mathbf{n},c}. \tag{21}$$

In terms of Schur basis matrix elements:

$$Q_{\mathbf{n},c} = \sum_\mu \langle s_\mu,\, \hat{Q}_{c,\mathbf{n}}\, s_\mu \rangle. \tag{22}$$

5 The adjoint Q-operator

5.1 Definition

Definition 10 (Adjoint Q-operator)**.** The adjoint is defined by:

$$\langle \hat{Q}^*_{c,\mathbf{n}} f,\ g\rangle = \langle f,\ \hat{Q}_{c,\mathbf{n}}\, g\rangle \qquad \text{for all } f, g \in \Lambda. \tag{23}$$

Since the (q)-Pochhammer factors are scalars:

$$\hat{Q}^*_{c,\mathbf{n}} = \prod_k (q;q)_{n_k} \cdot \hat{G}^*_{c,\mathbf{n}}. \tag{24}$$

The nontrivial content is in $\hat{G}^*_{c,\mathbf{n}}$, which inherits from $\hat{O}^*_c$ the profile transformation: the adjoint corresponds to the **reversed-complementary profile** $\bar{\pi}^{\mathrm{rev}}$ with reversed parameters.

5.2 Matrix elements: the transpose relation

In the Schur basis:

$$\langle s_\lambda,\ \hat{Q}^*_{c,\mathbf{n}}\, s_\mu\rangle = \langle s_\mu,\ \hat{Q}_{c,\mathbf{n}}\, s_\lambda\rangle. \tag{25}$$

The matrix of $\hat{Q}^*_{c,\mathbf{n}}$ is the *transpose* of the matrix of $\hat{Q}_{c,\mathbf{n}}$.

5.3 Trace invariance

Proposition 11. $\mathrm{tr}(\hat{Q}^*_{c,\mathbf{n}}) = \mathrm{tr}(\hat{Q}_{c,\mathbf{n}}) = Q_{\mathbf{n},c}$.

Proof.

$$\mathrm{tr}(\hat{Q}^*_{c,\mathbf{n}}) = \prod_k (q;q)_{n_k} \cdot \mathrm{tr}(\hat{G}^*_{c,\mathbf{n}}) = \prod_k (q;q)_{n_k} \cdot \mathrm{tr}(\hat{G}_{c,\mathbf{n}}) = Q_{\mathbf{n},c},$$

using Proposition 4 and the fact that scalar factors commute with the adjoint. □

Remark 12 (The adjoint is invisible at the polynomial level)**.** The adjoint $\hat{Q}^*_{c,\mathbf{n}}$ produces the same cylindric Kostka polynomial as $\hat{Q}_{c,\mathbf{n}}$. Its content is visible only at the level of individual matrix elements (the kernel).

6 The kernel and the positivity conjecture

6.1 The Q-kernel

Definition 13. The *Q-kernel* is:

$$K_{c,\mathbf{n}}(\lambda,\mu) := \langle s_\lambda,\ \hat{Q}_{c,\mathbf{n}}\, s_\mu\rangle\ :\ \ \mathrm{IP}\times\mathrm{IP} \to \mathbb{Q}[q]. \tag{26}$$

The adjoint kernel is the transpose: $K^*_{c,\mathbf{n}}(\lambda,\mu) = K_{c,\mathbf{n}}(\mu,\lambda)$. The diagonal is unchanged: $K^*(\mu,\mu) = K(\mu,\mu)$.

6.2 The positivity conjecture

Conjecture 14 (Multivariate positivity, operator form)**.** *For all* $\mathbf{n} \in \mathbb{N}^T$ *and all profiles* c *with* $d \not\equiv 0 \pmod 3$*:*

$$\mathrm{tr}(\hat{Q}_{c,\mathbf{n}}) = \sum_{\mu} K_{c,\mathbf{n}}(\mu,\mu) \in \mathbb{Z}_{\geq 0}[q]. \tag{27}$$

This is a statement about the **sum of diagonal matrix elements**. The conjecture does not require:

1. individual diagonal elements $K_{c,\mathbf{n}}(\mu,\mu)$ to be nonneg,
2. off-diagonal elements $K_{c,\mathbf{n}}(\lambda,\mu)$ to have any sign, or
3. self-adjointness ($\hat{Q} = \hat{Q}^*$).

6.3 The symmetrised operator

The *symmetrised* Q-operator

$$\hat{Q}^{\mathrm{sym}}_{c,\mathbf{n}} := \tfrac{1}{2}(\hat{Q}_{c,\mathbf{n}} + \hat{Q}^*_{c,\mathbf{n}}) \tag{28}$$

is self-adjoint by construction, with the same trace: $\mathrm{tr}(\hat{Q}^{\mathrm{sym}}) = Q_{\mathbf{n},c}$.

Its kernel is symmetric: $K^{\mathrm{sym}}(\lambda,\mu) = \frac{1}{2}(K(\lambda,\mu) + K(\mu,\lambda))$. When restricted to a finite-dimensional subspace of Λ, the eigenvalues of $\hat{Q}^{\mathrm{sym}}$ are real, and their sum equals $Q_{\mathbf{n},c}$.

Remark 15 (Positive semi-definiteness)**.** If $\hat{Q}^{\mathrm{sym}}$ were positive semi-definite (all eigenvalues ≥ 0), then $Q_{\mathbf{n},c} \geq 0$. This is strictly stronger than the conjecture (it implies each diagonal element is nonneg, not just their sum).

7 The commutation relations at Level 3

7.1 Vertex operator commutation

The fundamental commutation relation (Schur case) is:

$$\Omega^*[Xu]\,\Omega[Xv] = \frac{1}{1-uv}\,\Omega[Xv]\,\Omega^*[Xu]. \tag{29}$$

Each commutation produces a scalar factor $1/(1-uv) = \sum_{m\geq 0}(uv)^m$, generating the face labels of the growth diagram.

7.2 Degree operator commutation

$$D_w\,\Omega[Xu] = \Omega[Xuw]\,D_w, \tag{30}$$

$$D_w\,\Omega^*[Xu] = \Omega^*[Xu/w]\,D_w. \tag{31}$$

These are used in the proof of the Borodin product formula: commuting all Ω^* past all Ω produces scalar factors, and iterating the D_w shift produces the infinite product.

7.3 Self-commutation

Since $(\Omega[Xv])^* = \Omega^*[Xv]$, the fundamental relation can be read as:

$$(\Omega[Xv])^*\,\Omega[Xv] = \frac{1}{1-v^2}\,\Omega[Xv]\,(\Omega[Xv])^*. \tag{32}$$

The scalar factor $1/(1-v^2)$ measures the non-commutativity of $\Omega[Xv]$ with its own adjoint.

8 From multivariate to bivariate

The bivariate generating function $\mathrm{GK}_c(z,q) = \sum_{\mathfrak{c}} z^{\max(\mathfrak{c})} q^{|\mathfrak{c}|}$ is obtained from the multivariate one by the non-linear *max extraction*:

$$z_0^{|\mu^0|}\, z_1^{|\mu^1|}\, z_2^{|\mu^2|} \;\longmapsto\; z^{\max_k \mu_1^k}\, q^{|\mathfrak{c}|}.$$

This does not correspond to a specialisation of the z_k, so it does not lift to the operator level. The bivariate $Q_{n,c}(q)$ cannot be obtained from $\hat{Q}_{c,\mathbf{n}}$ by a simple specialisation. The chain of implications is:

$$\text{Conjecture 14 (multivariate)} \underset{\text{max projection}}{\Longrightarrow} \text{Bivariate positivity (Warnaar, Conj. 2.7).}$$

9 Summary of Part I

Object	Definition	Role
$\hat{O}_c$	$\prod_k G^{\pi_k}(u_k)\, D_w$	Level 3 operator on Λ
$\hat{O}_c^*$	$D_w \prod_k G^{\bar{\pi}_{T-1-k}}(u_{T-1-k})$	Reversed-complementary profile
$\hat{G}_c(\mathbf{z})$	$(w;w)_\infty \cdot \hat{O}_c(\mathbf{z})$	Corrected operator (E–H)
$\hat{G}_{c,\mathbf{n}}$	$[\mathbf{z}^{\mathbf{n}}]\,\hat{G}_c(\mathbf{z})$	Multi-index coefficient operator
$\hat{Q}_{c,\mathbf{n}}$	$\prod_k (q;q)_{n_k} \cdot \hat{G}_{c,\mathbf{n}}$	Q-operator (denominators cleared)
$\hat{Q}^*_{c,\mathbf{n}}$	$\prod_k (q;q)_{n_k} \cdot \hat{G}^*_{c,\mathbf{n}}$	Adjoint Q-operator
$K_{c,\mathbf{n}}(\lambda,\mu)$	$\langle s_\lambda, \hat{Q}_{c,\mathbf{n}}\, s_\mu\rangle$	Q-kernel
$Q_{\mathbf{n},c}$	$\sum_\mu K_{c,\mathbf{n}}(\mu,\mu) = \mathrm{tr}(\hat{Q}_{c,\mathbf{n}})$	Multivariate cylindric Kostka poly.

Key structural facts:

1. $\hat{O}_c^*$ corresponds to the reversed-complementary profile $\bar{\pi}^{\mathrm{rev}}$.
2. $\mathrm{tr}(\hat{Q}^*) = \mathrm{tr}(\hat{Q}) = Q_{\mathbf{n},c}$: the adjoint is invisible at the polynomial level.
3. The positivity conjecture is a statement about the trace (sum of diagonal matrix elements).
4. $\hat{Q}^{\mathrm{sym}} = \frac{1}{2}(\hat{Q} + \hat{Q}^*)$ is self-adjoint with the same trace; positive semi-definiteness would imply the conjecture but is strictly stronger.
5. The max extraction producing the bivariate $Q_{n,c}(q)$ does not lift to the operator level.

Part II
The Bivariate Q-Operator and Cascade Commutators

10 The operators P_m and T_n

We use the following mode operators on Λ:

- $P_m = h_m \cdot$: multiplication by the complete homogeneous symmetric function h_m. In generating function form, $P(z) = \sum_{m \geq 0} P_m z^m = \Omega[Xz]$.
- $T_n = P_n^* = \partial_{h_n}$: the adjoint of P_n with respect to the Hall inner product. In generating function form, $T(z) = \sum_{n \geq 0} T_n z^n$.

These satisfy the fundamental commutation relation (cf. [6], [2]):

$$T(v)\, P(w) = \frac{1}{1 - vw}\, P(w)\, T(v). \tag{33}$$

11 The bivariate $\mathbf{Q}(z)$ and $\mathbf{Q}^*(w)$

Definition 16. The *Q-operator* (bivariate version) for profile c is the formal power series in z with operator coefficients:

$$\mathbf{Q}(z) = \sum_{m \geq 0} Q_{m,c}(q)\, z^m\, P_m = \sum_{m \geq 0} Q_{m,c}(q)\, z^m\, h_m \cdot, \tag{34}$$

where $Q_{0,c}(q) = 1$.

The *adjoint Q-operator* is:

$$\mathbf{Q}^*(w) = \sum_{n \geq 0} Q_{n,c}(q)\, w^n\, T_n = \sum_{n \geq 0} Q_{n,c}(q)\, w^n\, \partial_{h_n}. \tag{35}$$

Remark 17. This is a fundamentally different construction from Part I. In Part I, the Q-operator $\hat{Q}_{c,\mathbf{n}}$ is a single operator whose *trace* gives the polynomial $Q_{\mathbf{n},c}$. Here, the polynomials $Q_m(q)$ appear as *coefficients* multiplying the Pieri operators P_m and their adjoints T_n.

12 Commutation relations in the h-basis

12.1 Mode-level commutation

Extracting the coefficient of $v^n w^m$ from (33):

Proposition 18.

$$T_n\, P_m = \sum_{k=0}^{\min(n,m)} P_{m-k}\, T_{n-k}. \tag{36}$$

Proof. From $T(v)\,P(w) = \frac{1}{1-vw}\,P(w)\,T(v)$, the right-hand side is

$$\Big(\sum_{k\geq 0}(vw)^k\Big)\Big(\sum_a P_a\,w^a\Big)\Big(\sum_b T_b\,v^b\Big) = \sum_{k,a,b} P_a\,T_b\,v^{k+b}\,w^{k+a}.$$

Setting $n = k+b$ and $m = k+a$ gives k ranging from 0 to $\min(n,m)$, with $a = m-k$ and $b = n-k$. □

Corollary 19. *The commutator is:*

$$[T_n, P_m] = \sum_{k=1}^{\min(n,m)} P_{m-k}\,T_{n-k}. \tag{37}$$

In particular:

$$[T_1, P_1] = \mathrm{id}, \tag{38}$$

$$[T_n, P_n] = P_{n-1}T_{n-1} + P_{n-2}T_{n-2} + \cdots + \mathrm{id}. \tag{39}$$

Remark 20 (Comparison with power sum basis)**.** In the power sum basis, $[\frac{1}{n}\partial_{p_n},\, p_m\cdot] = \delta_{n,m}$ (the Heisenberg algebra). The h-basis introduces *cascade terms* $P_{m-k}T_{n-k}$ for intermediate values of k. These cross-level terms are the key structural feature.

12.2 Lowering applied to h_m

Lemma 21.

$$T_n(h_m) = \begin{cases} h_{m-n} & \text{if } m \geq n, \\ 0 & \text{if } m < n. \end{cases} \tag{40}$$

Proof. $T_n(h_m) = T_nP_m(1) = \sum_{k=0}^{\min(n,m)} P_{m-k}T_{n-k}(1)$. Since $T_j(1) = 0$ for $j \geq 1$, only $k = n$ survives (when $n \leq m$), giving $P_{m-n}(1) = h_{m-n}$. □

13 Operator products applied to the vacuum

13.1 The product $\mathbf{Q}^*(w)\,\mathbf{Q}(z)$

Proposition 22.

$$\mathbf{Q}^*(w)\,\mathbf{Q}(z)\,[1] = \sum_{a\geq 0} C_a(zw)\,z^a\,h_a, \tag{41}$$

where $C_a(t) := \sum_{n\geq 0} Q_{a+n}(q)\,Q_n(q)\,t^n$ *is the cross-correlation function.*

Proof. By (34), $\mathbf{Q}(z)[1] = \sum_m Q_m z^m h_m$. Applying $\mathbf{Q}^*(w)$:

$$\begin{aligned} \mathbf{Q}^*(w)\big[\sum_m Q_m z^m h_m\big] &= \sum_{m,n} Q_m Q_n\, z^m w^n\, T_n(h_m) \\ &= \sum_{\substack{m,n \\ n\leq m}} Q_m Q_n\, z^m w^n\, h_{m-n} \qquad \text{(by Lemma 21)} \\ &= \sum_{a\geq 0} h_a\, z^a \sum_{n\geq 0} Q_{a+n}\,Q_n\,(zw)^n, \end{aligned}$$

substituting $a = m-n$. □

13.2 The product $\mathbf{Q}(z)\,\mathbf{Q}^*(w)$

Since $T_n(1) = 0$ for $n \geq 1$:

$$\mathbf{Q}(z)\,\mathbf{Q}^*(w)\,[1] = \mathbf{Q}(z)\,[1] = \sum_{m\geq 0} Q_m\, z^m\, h_m. \tag{42}$$

13.3 The commutator on the vacuum

Theorem 23.

$$[\mathbf{Q}(z),\, \mathbf{Q}^*(w)]\,[1] = -\sum_{a\geq 0} z^a\, h_a \sum_{n\geq 1} Q_{a+n}\, Q_n\, (zw)^n. \tag{43}$$

Proof. Subtracting (41) from (42):

$$\begin{aligned}[\mathbf{Q}(z), \mathbf{Q}^*(w)][1] &= \sum_a Q_a z^a h_a - \sum_a z^a h_a \sum_{n\geq 0} Q_{a+n} Q_n (zw)^n \\ &= \sum_a z^a h_a \big(Q_a - Q_a Q_0 - \textstyle\sum_{n\geq 1} Q_{a+n} Q_n (zw)^n\big).\end{aligned}$$

Since $Q_0 = 1$, the first two terms cancel. □

14 The vacuum expectation value

Taking the Hall inner product of (41) with 1 (extracting the h_0 coefficient):

Corollary 24.

$$\langle 1 \mid \mathbf{Q}^*(w)\,\mathbf{Q}(z) \mid 1\rangle = C_0(zw) = \sum_{n\geq 0} Q_n(q)^2\, (zw)^n. \tag{44}$$

Remark 25. If each $Q_n(q) \in \mathbb{N}[q]$ (the positivity conjecture), then $Q_n(q)^2 \in \mathbb{N}[q]$ and the vacuum expectation value is manifestly q-positive. The converse does not hold: $Q_n^2 \in \mathbb{N}[q]$ does not imply $Q_n \in \mathbb{N}[q]$.

15 The cross-correlation functions

Definition 26. The *cross-correlation functions* are:

$$C_a(t) = \sum_{n\geq 0} Q_{a+n}(q)\, Q_n(q)\, t^n \qquad (a \geq 0). \tag{45}$$

15.1 $d = 2$ (Rogers–Ramanujan)

For $c = (1, 1, 0)$, $Q_n = q^{n^2}$, so:

$$C_a(t) = q^{a^2} \sum_{n\geq 0} q^{2n^2+2an}\, t^n. \tag{46}$$

The vacuum expectation is the theta-like sum $C_0(t) = \sum_{n\geq 0} q^{2n^2}\, t^n$.

15.2 $d = 4$

For $c = (2, 2, 0)$, the first few polynomials are:

$$Q_0 = 1,$$
$$Q_1 = q^2 + q^3 + q^4 + q^5.$$

The vacuum expectation begins: $C_0(t) = 1 + (q^4 + 2q^5 + 3q^6 + 4q^7 + 3q^8 + 2q^9 + q^{10})\, t + \cdots$

16 The full commutator on Λ

The full commutator of the Q-operators acting on a general symmetric function $f \in \Lambda$ takes the form:

Theorem 27.

$$[\mathbf{Q}(z), \mathbf{Q}^*(w)] = -\sum_{a,b\geq 0} P_a\, T_b \sum_{k\geq 1} Q_{a+k}\, Q_{b+k}\, z^{a+k}\, w^{b+k}. \tag{47}$$

Proof. Using the commutation $T_n P_m = \sum_{j=0}^{\min(n,m)} P_{m-j} T_{n-j}$ from Proposition 18:

$$\begin{aligned}\mathbf{Q}^*(w)\, \mathbf{Q}(z) &= \sum_{m,n} Q_m Q_n\, z^m w^n\, T_n P_m \\ &= \sum_{m,n} Q_m Q_n\, z^m w^n \sum_{j=0}^{\min(n,m)} P_{m-j}\, T_{n-j}.\end{aligned}$$

Setting $a = m - j$, $b = n - j$, $k = j$:

$$\mathbf{Q}^*(w)\, \mathbf{Q}(z) = \sum_{a,b\geq 0} P_a\, T_b\; z^a w^b \sum_{k\geq 0} Q_{a+k}\, Q_{b+k}\, (zw)^k.$$

The $\mathbf{Q}(z)\, \mathbf{Q}^*(w)$ product has only the $k = 0$ term (since P_a and T_b act independently):

$$\mathbf{Q}(z)\, \mathbf{Q}^*(w) = \sum_{a,b\geq 0} Q_a\, Q_b\; z^a w^b\; P_a\, T_b.$$

Subtracting gives the commutator with the $k = 0$ terms cancelling. □

Remark 28 (Cascade structure)**.** The coefficient of $P_a T_b$ in the commutator is

$$-z^a w^b \sum_{k\geq 1} Q_{a+k}(q)\, Q_{b+k}(q)\, (zw)^k,$$

involving *shifted* cross-correlations. The presence of terms with $a \neq b$ is the key difference from the power sum basis formulation, where the commutator is diagonal: $[\sum Q_m z^m p_m, \sum Q_n w^n \frac{1}{n}\partial_{p_n}] = \sum_{n\geq 1} Q_n^2 (zw)^n$.

Remark 29 (Cross-level coupling and the involution)**.** The $P_a T_b$ terms with $a \neq b$ couple different depth levels of the cylindric partition. In the involution framework (see [4]), the involution must pair elements at level m with elements at level m'. The cascade terms provide the algebraic mechanism for such cross-level interactions.

Part III
The Generalized Local Rule

17 The standard algebra–bijection correspondence

We begin by recalling the precise dictionary between the algebraic framework (commutation relations) and the bijective framework (growth diagram local rules). This dictionary is the foundation on which the Q-enriched generalisation is built.

17.1 The Burge local rule as a commutation step

The fundamental commutation relation (33),

$$T(v)\,P(w) = \frac{1}{1-vw}\,P(w)\,T(v),$$

is *equivalent* to the Burge local rule at the mode level. Extracting the coefficient of $v^n w^m$ from both sides gives the cascade relation (Proposition 18):

$$T_n\,P_m = \sum_{k=0}^{\min(n,m)} P_{m-k}\,T_{n-k}. \tag{48}$$

Each term in the sum corresponds to one possible output of the local rule: given an input diamond $(\nu;\,\alpha,\beta)$ where $|\alpha|-|\nu| = m$ and $|\beta|-|\nu| = n$, the cascade parameter k determines the output $\lambda = \mathfrak{U}_{\alpha,\beta}(k,\nu)$ with $|\lambda| = |\alpha|+|\beta|-|\nu|+k$.

Remark 30 (The dictionary)**.** The correspondence between the algebraic proof and the bijective proof is summarised as follows (cf. [2], Remark 8):

Algebraic operation	**Bijective operation**
Commuting $T(v)$ past $P(w)$	Applying the local rule $\mathfrak{U}_{\alpha,\beta}$
Scalar factor $\frac{1}{1-vw} = \sum_k (vw)^k$	Face label $k \in \mathbb{Z}_{\geq 0}$
Cascade parameter k	Overlap $\lvert A \cap B\rvert$ in conjugate coordinates
Cyclic invariance of trace	Rotation of the cylinder
Commuting all T's past all P's	Full growth diagram computation
D_w shift: $D_w P(u) = P(uw) D_w$	Depth increment (adding base partition)

17.2 Face labels and the generating function

In the standard growth diagram for profile c, the face labels are non-negative integers $k \in \mathbb{Z}_{\geq 0}$, each contributing weight 1 regardless of the value of k. The face-label generating function for a single diamond is therefore:

$$\sum_{k\geq 0} t^k = \frac{1}{1-t}, \tag{49}$$

which is precisely the scalar factor arising from commuting $T(v)$ past $P(w)$ with $t = vw$. This is *universal*: it depends neither on the profile c nor on the polynomials $Q_n(q)$.

18 The Q-kernel

18.1 Derivation from the cascade commutator

The cascade commutator (Theorem 27) showed that the product $\mathbf{Q}^*(w)\,\mathbf{Q}(z)$ takes the form:

$$\mathbf{Q}^*(w)\,\mathbf{Q}(z) = \sum_{a,b\geq 0} P_a\,T_b\;z^a w^b \sum_{k\geq 0} Q_{a+k}\,Q_{b+k}\,(zw)^k. \tag{50}$$

This motivates extracting the inner sum as an independent object.

Definition 31 (Q-kernel)**.** The *Q-kernel* is the bivariate function:

$$\Phi_{a,b}(t) := \sum_{k\geq 0} Q_{a+k}(q)\,Q_{b+k}(q)\,t^k \qquad (a,b\geq 0), \tag{51}$$

where $Q_n = Q_{n,c}(q)$ are the cylindric Kostka polynomials for profile c.

With this notation, the cascade product becomes:

$$\mathbf{Q}^*(w)\,\mathbf{Q}(z) = \sum_{a,b\geq 0} P_a\,T_b\;z^a w^b\;\Phi_{a,b}(zw), \tag{52}$$

and the commutator (Theorem 27) isolates the $k\geq 1$ terms:

$$[\mathbf{Q}(z),\,\mathbf{Q}^*(w)] = -\sum_{a,b\geq 0} P_a\,T_b\;z^a w^b\;\big(\Phi_{a,b}(zw) - Q_aQ_b\big). \tag{53}$$

18.2 Relation to the cross-correlations

The Q-kernel unifies the cross-correlation functions from Section 15:

Proposition 32.

$$\Phi_{a,0}(t) = C_a(t) = \sum_{n\geq 0} Q_{a+n}\,Q_n\,t^n \qquad (a\geq 0). \tag{54}$$

In particular, $\Phi_{0,0}(t) = C_0(t) = \sum_{n\geq 0} Q_n^2\,t^n$ *is the vacuum expectation value.*

Proof. Set $b=0$ in (51) and use $Q_0 = 1$. □

18.3 Properties of the Q-kernel

Proposition 33. *The Q-kernel satisfies:*

1. Initial value. $\Phi_{a,b}(0) = Q_a\,Q_b$.
2. Symmetry. $\Phi_{a,b}(t) = \Phi_{b,a}(t)$.
3. Vacuum. $\Phi_{0,0}(t) = C_0(t)$.
4. Cross-correlation. $\Phi_{a,0}(t) = C_a(t)$.
5. Shifted factorisation. $\Phi_{a,b}(t) = Q_aQ_b + t\,\Phi_{a+1,b+1}(t)$.

Proof. Properties (1)–(4) follow directly from the definition. For (5), separate the $k = 0$ term:

$$\Phi_{a,b}(t) = Q_a Q_b + \sum_{k\geq 1} Q_{a+k}Q_{b+k}\, t^k = Q_a Q_b + t \sum_{k\geq 0} Q_{a+1+k}Q_{b+1+k}\, t^k = Q_a Q_b + t\, \Phi_{a+1,b+1}(t). \quad \square$$

Remark 34 (The Q-kernel as a deformation of $1/(1-t)$)**.** When $Q_n = 1$ for all n (the $d = 1$ case), $\Phi_{a,b}(t) = \sum_{k\geq 0} t^k = 1/(1-t)$, independent of a and b. For general profiles, $\Phi_{a,b}(t)$ is a *profile-dependent* deformation of this universal kernel. The shifted factorisation in property (5) can be iterated:

$$\Phi_{a,b}(t) = \sum_{j=0}^{N-1} Q_{a+j}\, Q_{b+j}\, t^j + t^N \Phi_{a+N,b+N}(t),$$

so the Q-kernel is determined by the sequence $(Q_{a+k}Q_{b+k})_{k\geq 0}$ in the same way that $1/(1-t)$ is determined by the constant sequence $(1, 1, 1, \ldots)$.

19 The generalised local rule

19.1 Definition

The standard Burge local rule assigns to each input diamond $(\nu;\, \alpha, \beta)$ and cascade parameter $k \geq 0$ an output partition $\lambda = \mathfrak{U}_{\alpha,\beta}(k, \nu)$ with weight $|\lambda| = |\alpha| + |\beta| - |\nu| + k$. In the growth diagram, the cascade parameter becomes the face label, and each label carries uniform weight 1.

The Q-enriched local rule uses the *same* partition geometry but assigns *non-uniform* weights to the cascade parameter.

Definition 35 (Q-enriched local rule)**.** Given an input diamond $(\nu;\, \alpha, \beta)$ with $|\alpha| - |\nu| = a$ and $|\beta| - |\nu| = b$, the *Q-enriched local rule* assigns to the cascade parameter k the weight:

$$\mathrm{wt}(k;\, a, b) = Q_{a+k}(q) \cdot Q_{b+k}(q). \tag{55}$$

The output partition is the same as in the standard rule: $\lambda = \mathfrak{U}_{\alpha,\beta}(k, \nu)$, with $|\lambda| = |\alpha|+|\beta|-|\nu|+k$.

The face-label generating function for the Q-enriched rule is therefore:

$$\sum_{k\geq 0} \mathrm{wt}(k;\, a, b)\, t^k = \sum_{k\geq 0} Q_{a+k}\, Q_{b+k}\, t^k = \Phi_{a,b}(t). \tag{56}$$

This replaces the universal $1/(1-t)$ of the standard rule with the profile-dependent Q-kernel.

Remark 36 (Weight depends on context)**.** In the standard rule, the face-label generating function $1/(1-t)$ is the same for every diamond, regardless of the strip sizes a and b. In the Q-enriched rule, the generating function $\Phi_{a,b}(t)$ depends on the strip sizes. This means the Q-enriched rule is *context-sensitive*: the weight assigned to a face label depends on the horizontal and vertical strips entering the diamond.

19.2 Weight preservation

The Q-enriched rule preserves the weight condition (48) of the standard rule.

Proposition 37. *Let $(\nu;\, \alpha, \beta)$ be an input diamond with $|\alpha| - |\nu| = a$ and $|\beta| - |\nu| = b$. For each cascade parameter $k \geq 0$, the output $\lambda = \mathfrak{U}_{\alpha,\beta}(k, \nu)$ satisfies:*

$$|\lambda| = |\alpha| + |\beta| - |\nu| + k = a + b + |\nu| + k. \tag{57}$$

This is a property of the Burge local rule itself and is independent of the weight assignment.

19.3 Invertibility

Proposition 38. *The Q-enriched local rule is invertible: given $(\lambda;\, \alpha, \beta)$, the cascade parameter k is uniquely determined by $k = |\lambda| - |\alpha| - |\beta| + |\nu|$, and the inner partition ν is recovered by the inverse Burge rule $\nu = \mathfrak{U}^{-1}_{\alpha,\beta}(k, \lambda)$.*

Proof. Invertibility is a property of the underlying Burge local rule (see [5], Proposition 2.20): the map $(\nu, k) \mapsto \lambda$ is a bijection from $\{(\nu, k) : \nu \in D(\alpha) \cap D(\beta),\ k \geq 0\}$ to $U(\alpha) \cap U(\beta)$. The Q-enriched rule uses the same bijection; only the weight assignment changes. □

20 The Q-enriched growth diagram

20.1 Standard growth diagrams and ALCDs

In the standard growth diagram for profile c, one fills a cylindric grid with partitions using the Burge local rule at each face. The face labels are non-negative integers, and the collection of all face labels forms an Arbitrarily Labelled Cylindric Diagram (ALCD). The key theorem (cf. [5], Theorem 2.21) is that this gives a weight-preserving bijection:

$$\psi_c \colon (\mathrm{IP},\, \mathrm{ALCD}(c)) \xrightarrow{\sim} \mathrm{CPP}(c), \tag{58}$$

where IP denotes the set of all integer partitions (the base partition γ). The multivariate generating function is:

$$\mathrm{GK}_c(\mathbf{z}) = \frac{1}{(w;\, w)_\infty} \cdot \sum_{\mathfrak{d} \in \mathrm{ALCD}(c)} \prod_{k=0}^{T-1} z_k^{|\mathrm{diag}(k)|_\mathfrak{d}}. \tag{59}$$

20.2 The Q-enriched growth diagram

Definition 39 (Q-ALCD)**.** A *Q-weighted Arbitrarily Labelled Cylindric Diagram* (Q-ALCD) is an ALCD $\mathfrak{d}$ together with the weight assignment:

$$\mathrm{wt}_Q(\mathfrak{d}) := \prod_{\text{faces } f} Q_{a_f + \mathfrak{d}(f)}(q) \cdot Q_{b_f + \mathfrak{d}(f)}(q), \tag{60}$$

where for each face f, the values a_f and b_f are the horizontal and vertical strip sizes entering that face.

The Q-enriched growth diagram uses the same partition geometry as the standard one—the same Burge local rule, the same set of valid ALCDs—but assigns the Q-dependent weight to each face rather than the uniform weight 1.

Remark 40 (Two levels of growth diagram)**.** This gives a natural hierarchy:

Level	**Face weight**	**Generating function**
Standard	1 (uniform)	Borodin product GK_c
Q-enriched	$Q_{a+k} \cdot Q_{b+k}$	Cross-correlations $\Phi_{a,b}$

The standard growth diagram produces the Borodin product formula for GK_c (summing uniformly over all ALCDs), while the Q-enriched growth diagram produces the cascade product $\mathbf{Q}^*(w)\, \mathbf{Q}(z)$, whose coefficients are the Q-kernels $\Phi_{a,b}(zw)$.

20.3 The commutator as obstruction

The commutator $[\mathbf{Q}(z),\, \mathbf{Q}^*(w)]$ from (53) measures the extent to which the Q-enriched creation ($\mathbf{Q}$) and annihilation ($\mathbf{Q}^*$) fail to commute:

$$[\mathbf{Q}(z),\, \mathbf{Q}^*(w)] = -\sum_{a,b\geq 0} P_a\, T_b\, z^a w^b\, \big(\Phi_{a,b}(zw) - Q_a Q_b\big). \tag{61}$$

The obstruction vanishes (the operators commute) if and only if $\Phi_{a,b}(t) = Q_a Q_b$ for all a, b, which happens precisely when $Q_n = 0$ for all $n \geq 1$. In the bijective framework, the commutator terms with $k \geq 1$ correspond to face labels that create "cross-level interactions" between different depths of the cylindric partition.

21 Special cases and examples

21.1 $d = 1$: recovery of the standard rule

For $d = 1$ (profiles $(1, 0, 0)$ and $(0, 1, 0)$), $Q_n(q) = 1$ for all n. The Q-kernel becomes:

$$\Phi_{a,b}(t) = \sum_{k\geq 0} 1 \cdot 1 \cdot t^k = \frac{1}{1-t}. \tag{62}$$

This is independent of a and b: every face carries the same generating function. The Q-enriched local rule reduces to the standard Burge local rule, confirming internal consistency.

21.2 $d = 2$ (Rogers–Ramanujan)

For $c = (1, 1, 0)$, $Q_n = q^{n^2}$ (the first Rogers–Ramanujan profile). The face weight for cascade parameter k is:

$$\mathrm{wt}(k;\, a, b) = q^{(a+k)^2 + (b+k)^2}. \tag{63}$$

The Q-kernel is:

$$\Phi_{a,b}(t) = \sum_{k\geq 0} q^{(a+k)^2+(b+k)^2}\, t^k = q^{a^2+b^2} \sum_{k\geq 0} q^{2k^2+2k(a+b)}\, t^k. \tag{64}$$

Example 41 ($\Phi_{0,0}$ and $\Phi_{1,0}$ for $d = 2$)**.**

$$\Phi_{0,0}(t) = \sum_{k\geq 0} q^{2k^2}\, t^k = 1 + q^2\, t + q^8\, t^2 + q^{18}\, t^3 + \cdots, \tag{65}$$

$$\Phi_{1,0}(t) = q \sum_{k\geq 0} q^{2k^2+2k}\, t^k = q + q^5\, t + q^{13}\, t^2 + \cdots. \tag{66}$$

The exponents $2k^2$ in $\Phi_{0,0}$ and $1 + 2k^2 + 2k = (2k+1)^2/2 + 1/2$ in $\Phi_{1,0}$ reflect the *quadratic form* structure underlying the Rogers–Ramanujan identities.

For the second profile $c = (2, 0, 0)$, $Q_n = q^{n(n+1)}$, and the face weight is $q^{(a+k)(a+k+1)+(b+k)(b+k+1)}$. The same quadratic structure appears.

21.3 $d = 4$, $c = (2, 2, 0)$

Here $Q_0 = 1$, $Q_1 = q^2 + q^3 + q^4 + q^5$, and the Q-kernel is no longer a monomial product.

Example 42 ($\Phi_{1,0}$ and $\Phi_{1,1}$ for $c = (2, 2, 0)$)**.**

$$\begin{aligned}\Phi_{1,0}(t) &= C_1(t) = Q_1 + Q_2\,Q_1\,t + Q_3\,Q_2\,t^2 + \cdots \\ &= (q^2+q^3+q^4+q^5) + Q_2(q^2+q^3+q^4+q^5)\,t + \cdots, \end{aligned} \tag{67}$$

$$\begin{aligned}\Phi_{1,1}(t) &= Q_1^2 + Q_2^2\,t + Q_3^2\,t^2 + \cdots \\ &= (q^4+2q^5+3q^6+4q^7+3q^8+2q^9+q^{10}) + Q_2^2\,t + \cdots. \end{aligned} \tag{68}$$

Note that $Q_1^2 = (q^2+q^3+q^4+q^5)^2$ is itself a polynomial with positive coefficients—this is a consequence of $Q_1 \in \mathbb{Z}_{\geq 0}[q]$ (confirmed by computation for this profile).

22 Connection to the involution

22.1 The peel/glue mechanism with Q-weights

The generalised local rule provides a concrete mechanism for the sign-reversing involution approach to proving positivity. Recall from [5] that the ground set $\mathcal{G}_{n,c}$ for the involution consists of pairs $(S, \mathfrak{d})$ where $S \subseteq \{1, \ldots, n\}$ with $|S| = j$ and $\mathfrak{d} \in \mathrm{ALCD}_{n-j}(c)$, with sign $(-1)^j$ and weight $q^{\sigma(S)+|\mathfrak{d}|}$.

In the Q-enriched framework, the face labels of $\mathfrak{d}$ carry the additional weight $\prod_f Q_{a_f+\mathfrak{d}(f)} \cdot Q_{b_f+\mathfrak{d}(f)}$. The involution must pair elements with opposite signs, and the unpaired elements (fixed points with $j = 0$) should contribute exactly $Q_{n,c}(q)$.

22.2 The Q-kernel as coupling constant

The Q-kernel $\Phi_{a,b}(t)$ quantifies the coupling between different depth levels. The $k = 0$ term $Q_a Q_b$ represents the "uncoupled" contribution (the product of independent Q-weights), while the terms with $k \geq 1$ represent *cross-level interactions*.

From (53), the commutator isolates precisely these cross-level terms:

$$\Phi_{a,b}(t) - Q_a Q_b = t\,\Phi_{a+1,b+1}(t) = \sum_{k\geq 1} Q_{a+k}\,Q_{b+k}\,t^k.$$

In the language of the involution: the terms with $k \geq 1$ are the contributions that the sign-reversing involution must cancel. The commutator provides the algebraic bookkeeping for these cancellations.

22.3 Non-negativity and the conjecture

Proposition 43. *If $Q_n(q) \in \mathbb{Z}_{\geq 0}[q]$ for all n (Warnaar's Conjecture 2.7), then for all a, b, k:*

$$Q_{a+k}(q) \cdot Q_{b+k}(q) \in \mathbb{Z}_{\geq 0}[q],$$

and consequently every coefficient of $\Phi_{a,b}(t)$ lies in $\mathbb{Z}_{\geq 0}[q]$. All face labels in the Q-enriched growth diagram carry non-negative q-weights.

This is immediate from the closure of $\mathbb{Z}_{\geq 0}[q]$ under multiplication, but it highlights an important structural point: the positivity conjecture ensures that the Q-enriched growth diagram has a combinatorial interpretation (all weights are non-negative). Conversely, if any Q_n has a negative coefficient, the corresponding face weights in the Q-enriched diagram become signed, and the combinatorial interpretation breaks down.

22.4 Open questions

1. **Connecting the two frameworks.** Part I defines $\hat{Q}_{c,\mathbf{n}}$ (an operator whose *trace* gives $Q_{\mathbf{n},c}$), while Part II defines $\mathbf{Q}(z)$ (an operator whose *coefficients* are the $Q_{n,c}$). Is there a natural relationship between $\hat{Q}_{c,\mathbf{n}}$ and the mode operators $Q_m P_m$?

2. **Positive semi-definiteness.** Can the positivity of $Q_{n,c}(q)$ be deduced from positive semi-definiteness of $\hat{Q}^{\mathrm{sym}}$ or from positive-definiteness of $\mathbf{Q}^*(w)\mathbf{Q}(z)$?

3. **Closed forms for $\Phi_{a,b}(t)$.** The Q-kernel $\Phi_{a,b}(t) = \sum Q_{a+k}Q_{b+k}\,t^k$ involves shifted products of Q-polynomials. For $d = 2$, it reduces to a theta-like sum (64). Does $\Phi_{a,b}(t)$ have a closed-form expression in terms of the Borodin product or other known q-series for general profiles?

4. **Matrix elements and the involution.** The matrix elements $\langle s_\lambda \mid P_a T_b \mid s_\mu \rangle$ count pairs of horizontal strips with common inner shape. Can these be used to construct the sign-reversing involution, using the Q-kernel as the coupling constant?

5. **The (r,t)-extension.** Both frameworks should extend to the Macdonald setting (replacing Schur by Macdonald polynomials, Pieri coefficients by $\varphi_{\lambda/\mu}(r,t)$ and $\psi_{\lambda/\mu}(r,t)$). Does the Q-kernel acquire additional structure in the (r,t)-deformed case?

6. **Explicit Q-enriched bijections.** Can the Q-enriched growth diagram be used to construct explicit bijections between sets of cylindric partitions weighted by Q-polynomials? In particular, does the bijection ψ_c from (58) extend to a Q-weighted version that directly produces the cross-correlations $\Phi_{a,b}$?

7. **The Q-kernel for $d \equiv 0 \pmod 3$.** For d divisible by 3, the polynomials Q_n are defined using $(q^\ell;\, q^\ell)_n$ with $\ell = 3$. The Q-kernel $\Phi_{a,b}(t)$ still makes sense formally, but the cyclic descent injection (see [4]) fails. Does the Q-kernel reveal why $d \equiv 0$ is special?

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